

KARA-SUU, Kyrgyzstan

WMO: 386160

Lat: 40.700N

Lon: 72.900E

Elev: 868

StdP: 91.32

Time Zone: 6.00 (E06)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.5	-9.1	-15.2	1.1	-10.3	-12.9	1.4	-8.4	5.2	3.6	3.9	-0.1	0.8	140	0.386

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.8	34.6	20.3	33.2	19.8	32.0	19.3	21.8	31.3	21.0	30.7	20.2	29.9	1.1	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.9	15.2	26.6	17.6	14.0	26.2	16.7	13.2	25.8	68.3	31.2	65.1	30.9	62.4	30.0	27.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.1	4.0	3.1	DB	-14.1	37.6	3.3	1.9	-16.5	39.0	-18.4	40.1	-20.3	41.2	-22.7	42.6	(4)
(5)			WB	-14.4	23.7	3.1	1.6	-16.7	24.8	-18.5	25.7	-20.2	26.6	-22.5	27.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	13.3	-0.6	1.8	8.7	14.6	19.2	23.5	25.6	24.4	20.3	13.5	6.7	0.8	(6)
(7)		DBStd	9.88	4.20	4.61	4.42	4.01	3.12	2.78	2.18	2.33	3.04	3.93	4.13	3.79	(7)
(8)		HDD10.0	1061	329	232	76	9	1	0	0	0	0	14	115	286	(8)
(9)		HDD18.3	2564	588	464	298	122	27	2	0	1	16	154	349	544	(9)
(10)		CDD10.0	2253	0	1	36	147	286	406	484	446	308	124	15	0	(10)
(11)		CDD18.3	713	0	0	0	9	53	158	226	188	74	5	0	0	(11)
(12)		CDH23.3	7393	0	0	1	87	513	1645	2468	1911	711	57	1	0	(12)
(13)		CDH26.7	3081	0	0	0	11	137	696	1178	844	207	7	0	0	(13)
(14)	Wind	WSAvg	1.1	0.9	1.1	1.3	1.3	1.3	1.1	1.1	1.1	1.0	0.9	0.8	(14)	
(15)	Precipitation	PrecAvg	401	34	44	54	50	50	24	12	7	29	45	45	(15)	
(16)		PrecMax	644	114	89	113	145	142	71	29	29	18	119	152	95	(16)
(17)		PrecMin	226	3	9	13	18	3	3	0	0	0	4	10	(17)	
(18)		PrecStd	96	22	20	27	26	36	20	9	7	6	26	37	21	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.9	16.3	22.7	28.3	31.6	35.6	36.5	35.1	31.6	27.5	21.6	12.0	(19)
(20)			MCWB	6.1	9.1	13.0	16.5	18.4	20.3	20.5	20.9	18.3	15.3	12.0	6.6	(20)
(21)		2%	DB	8.1	11.7	20.1	25.8	29.9	33.7	35.1	33.5	30.5	24.8	17.5	9.3	(21)
(22)			MCWB	4.5	6.9	11.9	15.5	18.0	19.6	20.4	20.1	17.7	14.4	10.5	5.4	(22)
(23)		5%	DB	6.1	9.6	17.7	23.7	28.2	32.2	33.6	32.4	29.2	22.8	14.9	7.4	(23)
(24)			MCWB	3.3	5.9	10.9	14.7	17.5	18.8	20.0	19.6	17.5	13.3	9.0	4.1	(24)
(25)		10%	DB	4.6	7.8	15.6	21.7	26.4	30.7	32.3	31.2	27.6	20.7	12.9	5.7	(25)
(26)			MCWB	2.3	4.9	9.8	13.9	16.7	18.4	19.7	19.2	16.7	12.4	8.0	3.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.8	9.7	14.5	18.3	20.2	22.6	23.3	22.7	20.6	16.3	13.0	7.7	(27)
(28)			MCDWB	10.5	15.2	19.8	25.3	28.9	32.0	32.5	32.2	29.4	25.6	20.1	10.5	(28)
(29)		2%	WB	4.9	7.5	12.6	16.7	19.1	20.9	21.9	21.6	19.1	15.0	11.1	5.8	(29)
(30)			MCDWB	7.3	10.5	18.4	23.4	27.7	30.5	31.6	30.9	28.4	23.1	16.1	8.4	(30)
(31)		5%	WB	3.6	6.2	11.3	15.5	18.2	19.8	21.1	20.7	18.0	14.0	9.7	4.5	(31)
(32)			MCDWB	5.7	9.1	16.4	21.9	26.2	29.9	31.2	30.0	27.1	21.2	14.0	6.9	(32)
(33)		10%	WB	2.5	5.0	10.3	14.5	17.3	19.0	20.4	20.0	17.0	13.0	8.6	3.4	(33)
(34)			MCDWB	4.3	7.5	14.8	20.5	24.9	28.8	30.4	29.3	26.2	19.2	12.3	5.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.0	7.3	8.9	10.0	11.5	12.9	12.8	12.7	12.5	11.1	8.6	7.0	(35)
(36)			MCDBR	8.5	10.0	12.3	13.1	14.3	14.9	14.6	13.8	14.1	13.5	11.8	9.3	(36)
(37)			MCWBR	6.0	6.3	6.5	6.7	6.6	6.2	6.2	6.0	6.6	6.5	6.6	6.4	(37)
(38)		5% WB	MCDWR	7.7	9.1	10.5	11.2	12.8	13.4	13.2	12.9	13.2	12.0	10.4	8.4	(38)
(39)			MCWBR	5.7	6.1	6.0	6.3	6.6	6.4	6.4	6.2	6.8	6.1	6.0	6.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.337	0.383	0.452	0.477	0.467	0.468	0.454	0.422	0.412	0.439	0.398	0.336	(40)	
(41)		taud	2.224	2.109	1.961	1.941	2.029	2.046	2.090	2.182	2.141	1.984	2.057	2.215	(41)	
(42)		Ebn at Noon	797	806	786	799	817	815	823	840	819	728	702	759	(42)	
(43)		Edh at Noon	98	127	166	181	169	167	158	140	136	143	115	91	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.23	3.10	4.21	5.19	6.26	7.15	7.28	6.69	5.49	3.57	2.35	1.90	(44)	
(45)		RadStd	0.16	0.24	0.31	0.40	0.41	0.32	0.30	0.28	0.22	0.36	0.30	0.17	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)		N/A	N/A	N/A	+0.51	N/A	N/A	N/A	N/A	+75	+64

Nomenclature: See separate page